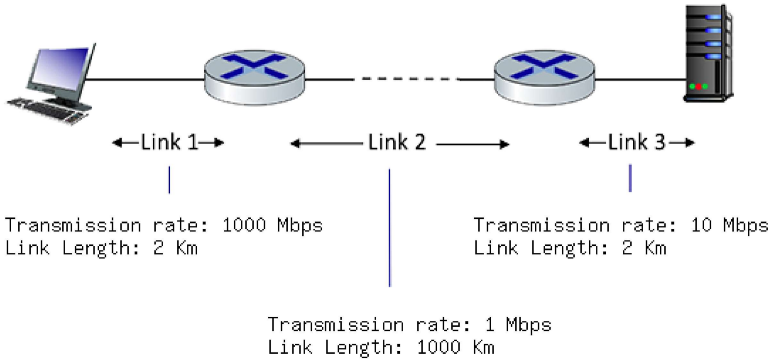
INTERACTIVE END-OF-CHAPTER EXERCISES

COMPUTING END-END DELAY (TRANSMISSION AND PROPAGATION DELAY)

Consider the figure below, with three links, each with the specified transmission rate and link length.



Assume the length of a packet is 4000 bits. The speed of light propagation delay on each link is 3x10^8 m/sec

Round your answer to two decimals after leading zeros

QUESTION LIST

1. What is the transmission delay of link 1?
2. What is the propogation delay of link 1?
3. What is the total delay of link 1?
4. What is the transmission delay of link 2?
5. What is the propogation delay of link 2?
6. What is the total delay of link 2?
7. What is the transmission delay of link 3?
8. What is the propogation delay of link 3?
9. What is the total delay of link 3?
10. What is the total delay?

SOLUTION

Link 1 transmission delay = L/R = 4000 bits / 1000 Mbps = 4.00E-6 seconds

Link 1 propagation delay = d/s = ()2 Km) * 1000 / 3*10^8 m/sec = 6.67E-6 seconds

Link 1 total delay = d_t + d_p = 4.00E-6 seconds + 6.67E-6 seconds = 1.07E-5 seconds

Link 2 transmission delay = L/R = 4000 bits / 1 Mbps = 0.004 seconds

Link 2 propagation delay = d/s = ()1000 Km) * 1000 / 3*10^8 m/sec = 0.0033 seconds

Link 2 total delay = d_t + d_p = 0.004 seconds + 0.0033 seconds = 0.0073 seconds

Link 3 transmission delay = L/R = 4000 bits / 10 Mbps = 0.0004 seconds

Link 3 propagation delay = d/s = ()2 Km) * 1000 / 3*10^8 m/sec = 6.67E-6 seconds

Link 3 total delay = d_t + d_p = 0.0004 seconds + 6.67E-6 seconds = 0.00041 seconds

The total delay = d_L1 + d_L2 + d_L3 = 1.07E-5 seconds + 0.0073 seconds + 0.00041 seconds = 0.0078 seconds

QUESTION 1 OF 10

What is the transmission delay of link 1?

Answer

CHECK HINT SKIP

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